

An operator-theoretic representation of the Weil quadratic form

Dylan Suvlu

24 September 2026

Abstract. The Weil explicit formula expresses a Hermitian pairing through prime powers and the Γ and polar terms of the completed ζ function; restricting it to a finite interval gives the localized Weil form. We represent this form as the boundary response of a completed arithmetic operator family. On the full line, the functional equation makes the squared output norm of each fixed source stationary at the critical parameter $s = \frac{1}{2}$. For a source supported in a fixed interval, we prove that the Weil form on its retained completed output is the negative derivative, evaluated at $s = \frac{1}{2}$, of the energy escaping the interval under real variation of s . The same pairing is half of a compressed difference of two orthogonal projections, $P_- - \mathcal{U}P_+\mathcal{U}^*$, for a unitary $\mathcal{U}$ constructed from the completed family.

The family is built from one-sided translations in logarithmic coordinates, which give finite Euler and Möbius identities on bounded windows. The uncompleted critical half-line operator is not closable. Adjoining the normalized Γ and polar factors, however, yields a bounded convolution operator whose explicit kernel is a weighted Müntz transform of a mean-zero polynomial. We construct a holomorphic family near the critical parameter. We then show that the critical source-Gram derivative on the half-line is trace class and is the trace-norm limit of its finite support compressions.

We also derive the complete localized Fourier matrix: the Γ multiplier on the diagonal plus divided differences of one explicit real odd function. When the full least eigenvalue is simple and its eigenvector is even, we recover the Connes–Consani–Moscovici finite characteristic, with only real zeros, as the Fourier transform of the ground state. The completed finite-window range consists precisely of H^1 functions vanishing at the terminal endpoint, and its projection onto every finite Fourier section is surjective. Positivity of the form and the limits of the characteristics as the interval grows are studied in companion papers.

1. Introduction

The Weil explicit formula expresses a Hermitian pairing in two ways: through prime-power translations together with the Γ and polar terms of the completion, and through evaluations at zeros of the completed ζ function [1, §3, (3.1)–(3.6)]

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-\frac{s}{2}}\Gamma\left(\frac{s}{2}\right)\zeta(s). \quad (1)$$

Restricting this pairing to a finite interval gives a localized form, and a further restriction to a finite Fourier space gives a Hermitian matrix. We seek an operator interpretation that explains both the arithmetic coefficients of this form and the effect of the interval boundary.

Finite restrictions also have a spectral significance. Under the hypothesis that the full least eigenvalue is simple and its eigenvector is even, the construction of Connes, Consani and Moscovici associates an entire characteristic with only real zeros to that eigenvector [1, Definition 5.3, Lemma 5.4 and Theorem 5.10]. This finite theorem does not by itself control

a family in which the interval and Fourier cutoff both grow. It motivates the study of the complete arithmetic matrix and of the operator structure from which its form is obtained.

In logarithmic coordinates, multiplication of integers becomes composition of one-sided translations. On a bounded interval, these translations give finite Euler and Möbius identities, and logarithmic differentiation produces the von Mangoldt coefficients. On the half-line, however, the uncompleted critical Euler operator is not closable. We show that adjoining the normalized Γ and polar factors produces a bounded convolution operator with an explicit integrable kernel. The completed operator belongs to a holomorphic family compatible with every finite support restriction.

The functional equation determines the critical variation of this family. On the full line, the squared output norm of each fixed source is stationary under real variation of the arithmetic parameter at one-half. For a source supported in a fixed interval, the output separates into the part retained in the interval and the part escaping to its left. Their squared-norm derivatives cancel. If these outputs are denoted by $X_L(s)f$ and $\mathcal{L}_L(s)f$, respectively, we prove

$$W_L\left(X_L\left(\frac{1}{2}\right)f, X_L\left(\frac{1}{2}\right)f\right) = -\left[\frac{d}{d\sigma}\|\mathcal{L}_L(\sigma)f\|_2^2\right]_{\sigma=\frac{1}{2}}. \quad (2)$$

Thus the localized Weil form on completed outputs is a boundary response of a globally stationary family. This identity does not determine its sign: a nonnegative escaped energy can have a derivative of either sign.

We also compute the complete Fourier matrix of the localized form, including its diagonal, and identify its relation to the finite real-zero construction. The completed finite-window range satisfies an absorbing boundary condition, whereas the Fourier characteristic uses periodic coordinates; projection makes every finite Fourier vector representable, but does not give uniform control of its source. The resulting framework separates construction of the completed arithmetic object from the further questions of response positivity and zero-retaining convergence of selected finite characteristics.

These two questions lead to separate companion branches. The positive-rate response is studied through θ correlations, one-sided causality and an actual Hankel-range criterion [2]; its supported arithmetic inverse and endpoint regularity are developed in [3]. The finite companion [4] instead studies the same Fourier matrix and selected characteristic. On one unbounded full simple-even selected family, a bounded complete characteristic mean gives compactness and zero retention. Its untouched lattice terms force the Fourier resolution required by that argument. That argument uses shifted positivity from the full finite minimum, independently of the positive-rate sign condition. Here $\Xi(z) = \xi(\frac{1}{2} + iz)$ denotes the function whose zero values are to be retained.

To state the operator theorem, let

$$(S_y f)(t) = f(t + y), \quad t, y \geq 0, \quad (3)$$

be the backward shift on $\mathcal{H}_+ = L^2(0, \infty)$. Its invariant subspace $\mathcal{H}_L = L^2(0, L)$ is understood by zero extension. On this subspace $S_y = 0$ for $y \geq L$. Consequently the operator

$$Z_L(s) = \sum_{n < e^L} n^{-s} S_{\log n} \quad (4)$$

is a finite sum for every complex s . It has a finite Möbius inverse, and its logarithmic derivative supplies exactly the von Mangoldt weights. Here completion means adjoining the normalized Γ and polar factors of the completed ζ function. It adds a smoothing real-place factor whose inverse is defined on a proper range.

On the dense domain $\mathcal{D}_c = \cup_{L>0} \mathcal{H}_L$, put

$$Z_+\left(\frac{1}{2}\right)f = \sum_{n \geq 1} n^{-\frac{1}{2}} S_{\log n} f. \quad (5)$$

Every such sum terminates. For real $b > 0$, let $R_b = \int_0^\infty e^{-by} S_y dy$, where the integral is applied to each vector. Define on $\mathcal{D}_c$

$$X_+^c = (I - 2R_{\frac{5}{2}})(I - 2R_{\frac{3}{2}})2\pi R_{\frac{1}{2}}Z_+\left(\frac{1}{2}\right). \quad (6)$$

The reference $s_0 = \frac{5}{2}$ makes the Γ -ratio exponent $\frac{s_0-s}{2}$ equal to 1 at $s = \frac{1}{2}$, so that its real-place factor reduces to $2\pi R_{\frac{1}{2}}$. The two preceding factors are the polar factors at the same reference; §2 derives both from the completed ζ factors.

Theorem 1 (Completed half-line operator). The operator in (6) extends uniquely to a bounded operator X_+ on $\mathcal{H}_+$. With $m = \lfloor e^y \rfloor$ and $\theta = e^y - m$, it is given by

$$X_+ f = \int_0^\infty K(y) S_y f dy, \quad K(y) = \pi e^{-\frac{5y}{2}} m(m+1)(1-2\theta), \quad (7)$$

and satisfies $\|X_+\| \leq \frac{8\pi}{3}$. Its restriction to $\mathcal{H}_L$ equals the critical completed finite-window operator defined in §2. The operator X_+ is not bounded below, whereas $Z_+(\frac{1}{2})$ on $\mathcal{D}_c$ is not closable.

The proof combines the Γ and polar factors before estimating their kernels. The leading integer-density terms cancel, leaving the fractional-part factor in (7). In multiplicative coordinates this kernel is a weighted Müntz transform of the mean-zero polynomial $2\pi(3t^2 - 2t)$ on $(0, 1]$. This connects the calculation with the sum-minus-integral operators of the Nyman–Beurling literature; see Báez-Duarte [5, §§2.2–2.3]. Section 3 proves the kernel formula and both operator obstructions. It also gives strong convergence of the physical support truncations.

The holomorphic family is defined on $|s - \frac{1}{2}| < \frac{1}{4}$. The escaped output in (2) is given by a Hilbert–Schmidt Hankel operator. Differentiating its Gram gives a trace-class operator on sources, whose finite support compressions converge in trace norm. The physical Fourier matrix below represents the form on Fourier outputs; identifying its vectors with completed sources additionally involves the range theorem.

The arithmetic-to-matrix step is made explicit in §4. We subtract the smooth density from the prime terms and combine it with the polar terms of the completed form. A real odd function then gives every off-diagonal entry through its divided differences. A separate diagonal calculation retains the original Γ multiplier and determines the derivatives of that function at the Fourier lattice. The resulting matrix represents the actual localized Weil form, without a ground-state hypothesis.

For the finite selection step, let E_N be the periodic Fourier polynomials of degree at most N on $(0, L)$, and let H_N represent the full localized Weil form on this space. Suppose its least eigenvalue ε_N is simple, with an eigenvector even under $t \mapsto L - t$. The displacement relation forces this eigenvector to have nonzero endpoint value. Thus there is a unique ground vector ξ_N with $\text{ev}_0 \xi_N = \xi_N(0) = 1$. With $D_N = -i \frac{d}{dt}$, the correction

$$D'_N = D_N - (D_N \xi_N) \text{ev}_0 \quad (8)$$

has the characteristic

$$\Theta_N(z) = \text{sinc}\left(\frac{Lz}{2}\right) \frac{\det(D'_N - zI)}{\det(D_N - zI)} = \frac{1}{L} \int_0^L \xi_N(t) e^{-iz(t-\frac{L}{2})} dt. \quad (9)$$

Section 5 proves this identity, including its extension at the free spectrum. Removing the ground null line makes D'_N self-adjoint for the positive metric induced by $H_N - \varepsilon_N I$, so the entire function in (9) has only real zeros.

This finite geometry belongs to Connes–Consani–Moscovici [1, §5], and to the more general distributional construction of Connes–van Suijlekom [6, Proposition 4.1 and Theorem 5.6]. The latter shows that the divided-difference and real-zero mechanism extends beyond the arithmetic distribution. Groskin [7, Corollary 2.4 and Theorem 2.5] also identifies an exact finite Guinand–Weil dictionary and its source quotient. The completed operator above realizes the arithmetic input before this finite spectral construction: the Hermitian pairing of its negative logarithmic derivative yields the form, whose lowest eigenline then determines the boundary characteristic. Throughout, the ground is the least eigenline of the full matrix; see also Connes’s survey [8, §6].

Suzuki’s screw-function approach gives a spatial primitive representation of the Weil form. The Weil distribution is the negative second derivative of a continuous function, and differentiation of test functions recovers its pairing [9, Proposition 3.1 and §3.5]. His subsequent operator treatment identifies a Friedrichs extension and constructs real-zero characteristics from self-adjoint derivative extensions in a shifted Weil metric [10, Theorems 1.1 and 1.5]. Here the completed one-sided arithmetic translations form a parameter family: for fixed sources and a fixed interval, the Weil form on completed outputs is the negative derivative of their escaped-energy pairing under real variation at the critical point.

Completion imposes an absorbing boundary condition: its finite-window range is exactly $\{f \in H^1(0, L) : f(L) = 0\}$. Its intersection with E_N therefore excludes ξ_N , although its orthogonal projection onto E_N is all of E_N . Section 6 gives this distinction, its finite spectral-gap estimate and an explicit inverse on the completed operator’s range. Quantitative preimage estimates would control transport through these source coordinates. The finite companion takes a different route: constant-mode transfer controls the filtered shifted energy without a uniform excitation gap [4, Theorems 1–3 and 5]. Its moving theorem assumes one unbounded full simple-even selected family with bounded complete characteristic mean; the Fourier cutoff is arbitrary, since the mean bound forces sufficient resolution. Those two family premises remain open; they give a sufficient zero-retaining entire cluster without requiring exact identification with normalized Ξ .

2. Arithmetic translations and completion

Throughout, inner products are conjugate-linear in the first argument. Fix $L > 0$, put $x = e^L$ and $\lambda = e^{\frac{L}{2}}$, and use Lebesgue measure on $\mathcal{H}_L$. Shifts on this space always mean killed backward shifts, not periodic translations. In this section all vector integrals take values in $\mathcal{H}_L$.

2.1. The finite shift algebra

For $a, b \geq 0$, direct substitution gives $S_a S_b = S_{a+b}$. Set $\mathcal{V}_n = S_{\log n}$ for positive integers n . Thus $\mathcal{V}_m \mathcal{V}_n = \mathcal{V}_{mn}$ and $\mathcal{V}_n = 0$ when $n \geq x$.

Proposition 2 (Euler synthesis and logarithmic derivative). For every $s \in \mathbb{C}$,

$$\begin{aligned} Z_L(s) &= \prod_{p < x} (I - p^{-s} \mathcal{V}_p)^{-1} = \sum_{n < x} n^{-s} \mathcal{V}_n, \\ Z_L(s)^{-1} &= \sum_{n < x} \mu(n) n^{-s} \mathcal{V}_n, \\ J_{p,L}(s) &= -Z_L(s)^{-1} Z'_L(s) = \sum_{1 < n < x} \Lambda(n) n^{-s} \mathcal{V}_n. \end{aligned} \tag{10}$$

Here p runs over primes, μ is the Möbius function and Λ is the von Mangoldt function. These are identities of bounded operators on $\mathcal{H}_L$.

We call $J_{p,L}(s)$ the prime current. For the completed operator, current will mean its negative logarithmic-derivative action on the domain justified below, after combining the arithmetic, Γ and polar factors.

Proof. Each $\mathcal{V}_p$ is nilpotent. Expanding each inverse as a terminating geometric series and using unique factorization proves the first equality. The product of the factors $I - p^{-s} \mathcal{V}_p$ has Möbius coefficients and is the inverse. All factors commute, so logarithmic differentiation gives the sum of $(\log p) p^{-ks} \mathcal{V}_{p^k}$ over prime powers. These are precisely the coefficients in the last line of (10). Every term with $\log n \geq L$ vanishes. $\square$

Finiteness here refers to the arithmetic sum; $\mathcal{H}_L$ is infinite-dimensional. A later Fourier projection need not preserve the product identity.

2.2. A Γ ratio on the absorbing domain

The absorbing endpoint also determines the generator domain. Its resolvent recovers a function by integrating back from that endpoint. The generator is

$$A_L f = f', \quad \text{Dom } A_L = \{f \in H^1(0, L) : f(L) = 0\}. \tag{11}$$

For every $b \in \mathbb{C}$, define

$$R_b f(t) = \int_t^L e^{-b(u-t)} f(u) du. \tag{12}$$

Differentiating this expression shows that $R_b = (b - A_L)^{-1}$, with range exactly $\text{Dom } A_L$. The triangular kernel is square-integrable; thus R_b is compact and injective. Fubini gives

$$R_b - R_c = (c - b) R_b R_c. \tag{13}$$

The inverse $b - A_L$ is unbounded in the ambient L^2 norm.

As in the introduction, choose $s_0 = \frac{5}{2}$ so that $\alpha = \frac{s_0 - s}{2}$ equals 1 at the critical point. For $\Re \alpha > 0$, set

$$E_{\infty,L}(s) = \frac{2\pi^\alpha}{\Gamma(\alpha)} \int_0^L e^{-sy} (1 - e^{-2y})^{\alpha-1} S_y dy. \tag{14}$$

The real positive base fixes the power. The scalar kernel is integrable at zero, and its parameter derivative has only an additional logarithmic singularity. It follows that (14) is a bounded operator, differentiable in operator norm locally in this parameter domain.

This factor is fixed by Euler's β integral, rather than by a prescribed matrix. For $\Re(s+r) > 0$ and $\Re\alpha > 0$, its corresponding scalar half-line integral is

$$\frac{2\pi^\alpha}{\Gamma(\alpha)} \int_0^\infty e^{-(s+r)y} (1 - e^{-2y})^{\alpha-1} dy = \frac{\Gamma_R(s+r)}{\Gamma_R(s_0+r)}, \quad (15)$$

where $\Gamma_R(w) = \pi^{-\frac{w}{2}} \Gamma(\frac{w}{2})$. Substitution $u = e^{-2y}$ proves the formula from the β integral [11, equation 5.12.1]. In particular, at $a = \frac{1}{2}$,

$$E_{\infty,L}(a) = 2\pi R_a. \quad (16)$$

Let $\mathcal{D}_0$ consist of bounded functions which are C^1 on each piece of a finite partition, with bounded derivatives and finite one-sided limits. They need not vanish at either endpoint. Splitting off intervals adjacent to the jumps and the terminal endpoint gives

$$\|S_y f - f\|_2 = O_f(\sqrt{y}), \quad f \in \mathcal{D}_0. \quad (17)$$

Consequently the following subtracted integral converges in $\mathcal{H}_L$:

$$J_{\Gamma,L} f = c f + \int_0^\infty \frac{e^{-ay} S_y f - e^{-2y} f}{1 - e^{-2y}} dy, \quad c = \frac{\gamma + \log \pi}{2}, \quad (18)$$

where γ is Euler's constant and $S_y = 0$ for $y \geq L$.

Lemma 3 (The range logarithmic derivative). For $f \in \mathcal{D}_0$,

$$E'_{\infty,L}(a) f = -E_{\infty,L}(a) J_{\Gamma,L} f. \quad (19)$$

In particular the derivative on the left belongs to $\text{Ran } E_{\infty,L}(a)$, so the expression $-E_{\infty,L}(a)^{-1} E'_{\infty,L}(a) f$ is defined and equals $J_{\Gamma,L} f$.

The subtraction in (18) is what permits differentiation at the operator level. We verify the identity with a lower cutoff before applying the range inverse.

Proof. Using $\psi(1) = -\gamma$ ([11, equation 5.4.12]), differentiation of (14) gives

$$E'_{\infty,L}(a) = 2\pi \int_0^L e^{-av} b(v) S_v dv, \quad b(v) = -v - c - \frac{1}{2} \log(1 - e^{-2v}). \quad (20)$$

Let J_ε be (18) with the integral restricted to $y \geq \varepsilon$. These are bounded operators, so convolution and Fubini apply to $R_a J_\varepsilon$. For $v > \varepsilon$, the coefficient of its kernel, divided by e^{-av} , is

$$c + \int_\varepsilon^v \frac{1}{1 - e^{-2y}} dy - \int_\varepsilon^\infty \frac{e^{-2y}}{1 - e^{-2y}} dy = c + v + \frac{1}{2} \log(1 - e^{-2v}) - \varepsilon. \quad (21)$$

For $0 < v < \varepsilon$ the coefficient is $c + \frac{1}{2} \log(1 - e^{-2\varepsilon})$. The latter interval contributes $O(\varepsilon(1 + |\log \varepsilon|))$ in kernel L^1 norm, while the error on (ε, L) is the constant $-\varepsilon$. Thus $R_a J_\varepsilon$ converges in operator norm to the convolution with coefficient $-b(v)$. By (17), $J_\varepsilon f$ converges to $J_{\Gamma,L} f$ for $f \in \mathcal{D}_0$. Combining these two limits and (16) proves (19). $\square$

For comparison with the localized explicit formula, put

$$\rho(y) = \frac{e^{\frac{y}{2}}}{e^y - e^{-y}}, \quad c_L = \gamma + \log\left(4\pi \tanh\left(\frac{L}{2}\right)\right). \quad (22)$$

Splitting the integral in (18) at L gives

$$J_{\Gamma,L}f = \left(\frac{c_L}{2}\right)f + \int_0^L \left(\rho(y)S_yf - \frac{f}{e^y - e^{-y}}\right) dy. \quad (23)$$

For example, the constant follows from

$$\int_0^L \frac{1}{e^y + 1} dy = \log\left(\frac{2}{1 + e^{-L}}\right), \quad \int_L^\infty \frac{1}{e^{2y} - 1} dy = -\frac{1}{2}\log(1 - e^{-2L}). \quad (24)$$

The two terms under the integral in (23) must be kept together.

2.3. Polar completion and the current

The polar polynomial supplies two linear factors, corresponding to its zeros at 0 and 1. Dividing each by its value at the reference s_0 gives the resolvent factors

$$\mathcal{P}_L(s) = [I + (s - s_0)R_{s_0}][I + (s - s_0)R_{s_0-1}]. \quad (25)$$

This realizes the scalar ratio $\frac{(s+r)(s+r-1)}{(s_0+r)(s_0+r-1)}$. It is the polar part of the conventional completion $\frac{1}{2}s(s-1)\Gamma_R(s)\zeta(s)$; see [11, equation 25.4.4]. The resolvent identity proves

$$[I + (s - s_0)R_{s_0-j}]^{-1} = I + (s_0 - s)R_{s-j}, \quad j = 0, 1. \quad (26)$$

Thus $\mathcal{P}_L(s)$ is boundedly invertible for every complex s . Another use of (13) yields

$$-\mathcal{P}_L(s)^{-1}\mathcal{P}'_L(s) = -R_s - R_{s-1}. \quad (27)$$

Define $X_L(s) = \mathcal{P}_L(s)E_{\infty,L}(s)Z_L(s)$ for $\Re s < \frac{5}{2}$. We call an input u a source for its completed output $X_L(s)u$. At $a = \frac{1}{2}$, its critical factors are those of (6) restricted to the interval. They commute as bounded shift convolutions. The factors $\mathcal{P}_L(a)$ and its inverse preserve $\text{Dom } A_L$, while $Z_L(a)$ is onto.

Hence

$$X_L(a) \text{ is injective, } \text{Ran } X_L(a) = \text{Dom } A_L. \quad (28)$$

On $\mathcal{D}_0$ put

$$J_L = J_{p,L}(a) + J_{\Gamma,L} - R_a - R_{a-1}. \quad (29)$$

By the product rule, (19) and (27),

$$X'_L(a)f = -X_L(a)J_Lf, \quad f \in \mathcal{D}_0. \quad (30)$$

To justify the product rule on this test domain, note that finite shifts preserve $\mathcal{D}_0$. The regularized Γ currents commute with these shifts; their limits therefore commute on the stated test domain. All remaining commutations involve bounded integral operators. In particular (30) proves range membership before the inverse of $X_L(a)$ is used.

The derivative also has a useful form in which the current acts on the completed output. Completion gains the regularity that this current requires.

For $f \in H^1(0, L)$, the interior translation estimate and the terminal strip give

$$\|S_yf - f\|_2 \leq y\|f'\|_2 + \sqrt{y}\|f\|_\infty \leq C_L\sqrt{y}\|f\|_{H^1}, \quad 0 < y < \min(1, L). \quad (31)$$

Thus (18) defines a continuous map $J_L : H^1(0, L) \rightarrow \mathcal{H}_L$. If $J_{L,\varepsilon}$ cuts off only its Γ integral below ε , then

$$\|J_{L,\varepsilon} - J_L\|_{H^1 \rightarrow L^2} \leq C_L \sqrt{\varepsilon}. \quad (32)$$

Each cutoff current is bounded on $\mathcal{H}_L$ and commutes with $X_L(a)$. The kernel calculation proving (19), followed by the bounded prime and polar product rule, gives

$$X_L(a)J_{L,\varepsilon} \rightarrow -X'_L(a) \quad \text{in operator norm.} \quad (33)$$

On the other hand, $X_L(a)$ maps $\mathcal{H}_L$ boundedly into $\text{Dom } A_L$ with its H^1 norm. This follows from $R_a : L^2 \rightarrow H^1$ and $A_L R_b = bR_b - I$, which also proves that the polar factors preserve this domain boundedly. Hence the cutoff currents composed on the left converge in operator norm as well. Their commutation proves

$$J_L X_L(a)u = -X'_L(a)u, \quad u \in \mathcal{H}_L. \quad (34)$$

This argument applies to every source u in L^2 ; it requires current regularity of the completed output.

We shall use the continuous Hermitian form on $H^1(0, L)$

$$W_L(f, g) = -\langle f, J_L g \rangle - \langle J_L f, g \rangle. \quad (35)$$

We call this negative Hermitian pairing the Hermitian current form. Section 4 identifies it with the localized Weil form. That comparison is first made on smooth interval vectors and extends to H^1 by continuity. In the subtracted integrals, the translation estimate above supplies an integrable $O(y^{-\frac{1}{2}})$ majorant at the origin. This form definition avoids any appeal to the domain of an unbounded adjoint.

3. Completion on the half-line and localization

3.1. The critical kernel

The kernel calculation proves Theorem 1 by combining the polar resolvents with the critical Γ factor before applying the arithmetic sum. The resulting sums of integers and squares cancel their leading density terms. Only then do we estimate the remaining fractional-part kernel.

For $b > 0$, the half-line R_b is bounded with norm at most $\frac{1}{b}$, and preserves every $\mathcal{H}_L$. On $\mathcal{D}_c$, resolvent algebra gives

$$(I - 2R_{\frac{5}{2}})R_{\frac{1}{2}} = R_{\frac{5}{2}}, \quad R_{\frac{5}{2}}(I - 2R_{\frac{3}{2}}) = 3R_{\frac{5}{2}} - 2R_{\frac{3}{2}}. \quad (36)$$

Therefore

$$X_+^c = 2\pi(3R_{\frac{5}{2}} - 2R_{\frac{3}{2}})Z_+\left(\frac{1}{2}\right). \quad (37)$$

For f supported in $(0, L)$, finite summation and Fubini give

$$R_b Z_+\left(\frac{1}{2}\right)f = \int_0^L e^{-by} \left(\sum_{n \leq e^y} n^{b-\frac{1}{2}} \right) S_y f \, dy. \quad (38)$$

Insert $b = \frac{5}{2}$ and $b = \frac{3}{2}$ into (37) and sum the integers and their squares. With $m = \lfloor e^y \rfloor$ the resulting kernel is

$$2\pi \left(3e^{\frac{-5y}{2}} \frac{m(m+1)(2m+1)}{6} - 2e^{\frac{-3y}{2}} \frac{m(m+1)}{2} \right) = \pi e^{\frac{-5y}{2}} m(m+1)(1 - 2\{e^y\}). \quad (39)$$

Its values at integer thresholds are immaterial to the integral. Since $m \leq e^y$ and $|1 - 2\{e^y\}| \leq 1$,

$$|K(y)| \leq \pi \left(e^{-\frac{y}{2}} + e^{-\frac{3y}{2}} \right), \quad \int_0^\infty |K(y)| dy \leq \frac{8\pi}{3}. \quad (40)$$

The vector integral with this kernel defines a bounded operator on all of $\mathcal{H}_+$. It agrees with X_+^c on its dense domain and hence is the unique extension. Each $\mathcal{H}_L$ is invariant, and all factors restrict to their finite-window definitions. This proves the asserted exact restrictions. Finally Q_L converges strongly to the identity, so boundedness implies $Q_L X_+ Q_L f \rightarrow X_+ f$ for every $f \in \mathcal{H}_+$.

The same cancellation can be written in the multiplicative variable $u = e^{-y}$. Let

$$h(t) = 2\pi(3t^2 - 2t)\mathbf{1}_{(0,1]}(t). \quad (41)$$

Then $\int_0^\infty h(t) dt = 0$, and direct summation gives

$$K(y) = \sqrt{u} \sum_{n \geq 1} h(nu). \quad (42)$$

Thus K is the weighted Müntz transform associated with this seed: the usual term $u^{-1} \int h$ is zero. Báez-Duarte [5, §2.3] studies the sum-minus-integral construction and its fractional-part representation. The seed has a jump at 1 and has bounded variation. Burnol [12, Proposition 3.9] treats the Müntz formula under this hypothesis; the preceding finite sum gives the present kernel explicitly.

The two operator obstructions.

The uncompleted synthesis cannot be extended by taking its graph closure. To see this, fix $d > 0$ and let

$$f_T(t) = e^{-\frac{T}{2}} \mathbf{1}_{[T, T+d]}(t). \quad (43)$$

Then $\|f_T\|^2 = de^{-T} \rightarrow 0$. For $0 \leq t \leq T$,

$$Z_+ \left(\frac{1}{2} \right) f_T(t) = e^{-\frac{T}{2}} \sum_{e^{T-t} \leq n \leq e^{T+d-t}} n^{-\frac{1}{2}}. \quad (44)$$

Integral comparison for the decreasing function $u^{-\frac{1}{2}}$ gives, uniformly for $b \geq a \geq 1$,

$$\sum_{a \leq n \leq b} n^{-\frac{1}{2}} = 2(\sqrt{b} - \sqrt{a}) + O(a^{-\frac{1}{2}}). \quad (45)$$

Consequently the last displayed synthesis differs from

$$2(e^{\frac{d}{2}} - 1)e^{-\frac{t}{2}} \quad (46)$$

by $O(e^{-T}e^{\frac{t}{2}})$ on $(0, T)$. The squared error integral is $O(e^{-T})$. On $(T, T+d)$ only integers $n \leq e^d$ occur, so its squared integral is $O_d(e^{-T})$; beyond $T+d$ the synthesis is zero. The tail of the proposed limit has the same vanishing bound. Hence

$$Z_+\left(\frac{1}{2}\right)f_T \rightarrow 2\left(e^{\frac{d}{2}} - 1\right)e^{-\frac{t}{2}} \neq 0 \text{ in } \mathcal{H}_+. \quad (47)$$

This proves nonclosability.

Completion annihilates this particular graph defect. Indeed $R_{\frac{3}{2}}e^{-\frac{t}{2}} = \left(\frac{1}{2}\right)e^{-\frac{t}{2}}$, so $I - 2R_{\frac{3}{2}}$ kills the limiting output. The factorization (37) remains an identity on $\mathcal{D}_c$; it is not an everywhere-defined factorization through a closed uncompleted arithmetic operator.

To finish Theorem 1, take the unit vectors $g_r(t) = \sqrt{2r}e^{-rt}$, $r > 0$. They are simultaneous eigenvectors of every backward shift, and

$$X_+g_r = k(r)g_r, \quad k(r) = \int_0^\infty K(y)e^{-ry} dy. \quad (48)$$

By (40) and dominated convergence, $k(r) \rightarrow 0$ as $r \rightarrow \infty$. Thus X_+ has no positive lower bound in the ambient norm. This completes the proof. $\square$

3.2. A holomorphic completed family

The cancellation behind (7) persists when the arithmetic parameter varies. The resulting kernels acquire integrable singularities at arithmetic thresholds, so a pointwise exponential bound is no longer the right estimate. We control them in a norm that measures both the forward operator and the output lost at a boundary.

Put $a = \frac{1}{2}$, $D = \{s \in \mathbb{C} : |s - a| < \frac{1}{4}\}$ and

$$\mathcal{B} = L^1(0, \infty) \cap L^2((0, \infty), (1 + y) dy). \quad (49)$$

Equip this intersection with the sum of its two norms. For a kernel in $\mathcal{B}$, backward convolution defines operators on both the half-line and the full line. We denote them by $X_+(s)$ and $\mathbb{X}(s)$, respectively.

Theorem 4 (Compatible parameter completion). There is a holomorphic map $s \mapsto k_s$ from D to $\mathcal{B}$ such that

$$X_+(s)f = \int_0^\infty k_s(y)S_y f dy, \quad (\mathbb{X}(s)f)(t) = \int_0^\infty k_s(y)f(t + y) dy. \quad (50)$$

Both operator families are holomorphic in operator norm. Every $\mathcal{H}_L$ is invariant under $X_+(s)$, and $X_+(s)|_{\mathcal{H}_L} = X_L(s)$. At a , $k_a = K$ almost everywhere. The kernel and every fixed parameter derivative are bounded in $\mathcal{B}$ uniformly on each smaller closed disk in D .

The construction uses one mean-zero seed rather than a separately chosen interpolation of the critical operator. With $\alpha = \frac{\frac{5}{2}-s}{2}$ and $c_\alpha = \frac{2\pi^\alpha}{\Gamma(\alpha)}$, set, for $0 < u < 1$,

$$h_\alpha(u) = c_\alpha \left[u^2(1 - u^2)^{\alpha-1} - 2\alpha u^{2\alpha-1} \int_u^1 t^{2-2\alpha}(1 - t^2)^{\alpha-1} dt \right]. \quad (51)$$

The powers use the real logarithm of their positive bases. Applying the two polar factors to the Γ kernel gives this seed, and Fubini gives $\int_0^1 h_\alpha(u) du = 0$. The arithmetic kernel is

$$k_s(y) = e^{-sy} \sum_{1 \leq n < e^y} h_\alpha(ne^{-y}), \quad (52)$$

away from $e^y \in \mathbb{N}$, where we choose the value zero. These countably many values have no effect on any operator. At $\alpha = 1$, the seed is $2\pi(3u^2 - 2u)$, recovering (7). Appendix B proves the seed calculation, the threshold estimates, holomorphy and the exact finite restrictions.

The mean-zero identity removes the continuum term in the arithmetic Riemann sum; the remaining threshold errors are summable in both norms of $\mathcal{B}$.

3.3. Stationarity and the escaped output

The critical completed operator now belongs to a holomorphic family near $s = \frac{1}{2}$. On the full line, the functional equation makes the squared output norm stationary at the critical point for every fixed input. Restricting that input to a finite interval separates the output into a retained part and an escaped part. We identify the derivative of the retained squared norm with the localized Weil form evaluated on the completed output.

Use the Fourier convention $\hat{f}(\omega) = \int_{\mathbb{R}} f(t)e^{-i\omega t} dt$. The full-line multiplier is $m_s(-i\omega)$, where

$$m_s(w) = \frac{2\xi(s+w)}{(\frac{5}{2}+w)(\frac{3}{2}+w)\Gamma_R(\frac{5}{2}+w)}, \quad \xi(z) = \left(\frac{1}{2}\right)z(z-1)\Gamma_R(z)\zeta(z). \quad (53)$$

Appendix B derives this formula first in a half-plane of absolute convergence and then takes its boundary value. In particular its reference denominator is fixed as s varies. The functional equation $\xi(z) = \xi(1-z)$ and real type give, for real σ in D ,

$$|\xi(\sigma - i\omega)| = |\xi(1 - \sigma - i\omega)|. \quad (54)$$

By Plancherel, $\mathbb{X}(\sigma)^*\mathbb{X}(\sigma) = \mathbb{X}(1-\sigma)^*\mathbb{X}(1-\sigma)$. Real differentiation at a yields

$$\left[\left(\frac{d}{d\sigma} \right) (\mathbb{X}(\sigma)^*\mathbb{X}(\sigma)) \right]_{\sigma=a} = 0. \quad (55)$$

Here and below a dot denotes this real derivative at a ; the holomorphic derivative of X has the same value along the real axis. The Gram itself is being differentiated as a real-parameter family.

Let $i_L : \mathcal{H}_L \rightarrow L^2(\mathbb{R})$ be zero extension and $P_L = i_L i_L^*$. Extend an interval source by zero and apply the full-line completed operator. We call the part of its output outside the interval boundary leakage. Because backward convolution samples to the right, this escaped output lies only to the left of the interval. This one-sided orientation is the meaning of causal below. The exterior-output map is

$$\mathcal{L}_L(s) = (I - P_L)\mathbb{X}(s)i_L. \quad (56)$$

It vanishes for $t > L$. Reflecting its remaining support by $t = -r$ identifies it with the Hankel operator

$$(\mathcal{L}_L(s)u)(r) = \int_0^L k_s(r+q)u(q) dq, \quad r > 0. \quad (57)$$

This identification is an isometry of the output space.

For fixed sources, $X_L(s)^*X_L(s)$ records the inner products of their completed outputs. We call this positive operator the source Gram and write it as $\mathcal{G}_L(s)$.

Theorem 5 (The Weil form as a leakage derivative). Fix $L > 0$. For $s \in D$, the maps $\mathcal{L}_L(s)$ are holomorphic with values in the Hilbert–Schmidt operators, and

$$\|\mathcal{L}_L(s)\|_{\text{HS}}^2 = \int_0^\infty \min(L, y) |k_s(y)|^2 dy. \quad (58)$$

Put $\mathcal{G}_L(s) = X_L(s)^* X_L(s)$, $X = X_L(a)$ and $\dot{X} = X'_L(a)$. Its real critical derivative is

$$\dot{\mathcal{G}}_L = \dot{X}^* X + X^* \dot{X}. \quad (59)$$

For every pair of fixed sources $u, v \in \mathcal{H}_L$,

$$W_L(Xu, Xv) = \langle u, \dot{\mathcal{G}}_L v \rangle = -\langle \dot{\mathcal{L}}_L u, \mathcal{L}_L v \rangle - \langle \mathcal{L}_L u, \dot{\mathcal{L}}_L v \rangle, \quad (60)$$

where the leakage factors on the right are evaluated at a . The operator $\dot{\mathcal{G}}_L$ is trace class, with a trace-norm bound independent of L .

Proof. Tonelli with $y = r + q$ proves (58). The map from a kernel in $L^2(y dy)$ to the displayed Hankel operator is bounded and linear; hence it preserves holomorphy and bounds every fixed derivative on smaller disks uniformly in L .

The two parts of the full-line output are orthogonal, and the retained part is exactly $X_L(s)$. Thus

$$X_L(s)^* X_L(s) = i_L^* \mathbb{X}(s)^* \mathbb{X}(s) i_L - \mathcal{L}_L(s)^* \mathcal{L}_L(s). \quad (61)$$

Differentiate for real s at a and use (55):

$$\dot{\mathcal{G}}_L = -\dot{\mathcal{L}}_L^* \mathcal{L}_L - \mathcal{L}_L^* \dot{\mathcal{L}}_L. \quad (62)$$

Products of two Hilbert–Schmidt maps are trace class, with trace norm bounded by the product of their Hilbert–Schmidt norms. Equation (58) applied also to k'_a gives the asserted uniform bound.

Finally (28) puts Xu, Xv in H^1 , and the all-source identity (34) gives $J_L Xu = -\dot{X}u$ and $J_L Xv = -\dot{X}v$. Substitution into (35) proves the first equality in (60). The second is (62). In particular,

$$W_L(Xu, Xu) = -\left[\left(\frac{d}{d\sigma} \right) \|\mathcal{L}_L(\sigma)u\|^2 \right]_{\sigma=a}. \quad (63)$$

The source and the interval are held fixed in this derivative. $\square$

The localized form on a completed output is therefore the negative derivative of the escaping source energy. The prime, Γ and polar terms of the localized explicit formula together measure the first-order change of the retained squared norm as the real arithmetic parameter varies at the fixed boundary. The full-line norm remains stationary: its retained and escaped parts have opposite derivatives. Positivity of the escaping energy does not determine the sign of either derivative.

3.4. A projection defect for completed outputs

The same boundary identity can be written using the critical operator and its parameter derivative alone. The resulting unitary also appears in the support criterion of the θ -correlation companion [2, §5]. We construct it here directly from the completed family.

Put $A_\pm = \mathbb{X}(a) \pm \dot{\mathbb{X}}(a)$, where the dot is the real parameter derivative at $a = \frac{1}{2}$. Equation (55) gives

$$A_+^* A_+ = A_-^* A_- . \quad (64)$$

To identify their ranges and orientation, set

$$\chi(\omega) = \frac{\xi(\frac{1}{2} + i\omega)}{\xi(\frac{1}{2})}, \quad d(\omega) = \frac{2\xi(\frac{1}{2})}{(\frac{5}{2} - i\omega)(\frac{3}{2} - i\omega)\Gamma_R(\frac{5}{2} - i\omega)}. \quad (65)$$

The multipliers of $\mathbb{X}(a)$ and $\dot{\mathbb{X}}(a)$ are respectively $d\chi$ and $id\chi'$. Indeed χ is real and even on the real line, and (53) is evaluated at $w = -i\omega$. Thus $A_\pm$ have multipliers $d(\chi \pm i\chi')$, which are nonzero almost everywhere: a common zero of χ, χ' belongs to the discrete zero set of the nonzero entire function χ . Their ranges are dense.

The equal-Gram identity makes $A_+ f \mapsto A_- f$ an isometry between these dense ranges. It extends uniquely to a unitary $\mathcal{U}$, whose multiplier is

$$S(\omega) = \frac{\chi(\omega) - i\chi'(\omega)}{\chi(\omega) + i\chi'(\omega)}. \quad (66)$$

This is the same unitary and Fourier orientation as in [2, §5]; no bounded inverse of A_+ is used.

Corollary (Completed-output projection defect). Let P_- be projection onto the negative physical half-line and put

$$\Delta_{\mathcal{U}} = P_- - \mathcal{U}P_- \mathcal{U}^*. \quad (67)$$

For every fixed $L > 0$ and sources $f, g \in \mathcal{H}_L$,

$$W_L(X_L(a)f, X_L(a)g) = \frac{1}{2} \langle A_- i_L f, \Delta_{\mathcal{U}} A_- i_L g \rangle. \quad (68)$$

Here $\Delta_{\mathcal{U}}$ is a self-adjoint difference of orthogonal projections.

Proof. For any orthogonal projection Q , expansion gives

$$\dot{\mathbb{X}}(a)^* Q \mathbb{X}(a) + \mathbb{X}(a)^* Q \dot{\mathbb{X}}(a) = \frac{1}{2} (A_+^* Q A_+ - A_-^* Q A_-). \quad (69)$$

Since $\mathcal{U}A_+ = A_-$, the negative of this expression is

$$\frac{1}{2} A_-^* (Q - \mathcal{U}Q\mathcal{U}^*) A_-. \quad (70)$$

For a source supported in $(0, L)$, backward convolution has no output beyond L , so its escaped part is exactly the negative-half-line output. Set $Q = P_-$ and apply (60) to the fixed zero-extended sources. This proves (68). $\square$

The identity represents the form on completed outputs. A sign condition on the projection difference is an additional support question, treated in the companion; the difference need not itself be a projection.

3.5. A trace-class limit in source coordinates

The same Hankel kernel acts on the entire positive half-line:

$$(\mathcal{K}(s)u)(r) = \int_0^\infty k_s(r+q)u(q) \, dq, \quad \|\mathcal{K}(s)\|_{\text{HS}}^2 = \int_0^\infty y |k_s(y)|^2 \, dy. \quad (71)$$

Put $\mathcal{G}_+(s) = X_+(s)^* X_+(s)$. Write $\mathcal{K} = \mathcal{K}(a)$ and $\dot{\mathcal{K}} = \mathcal{K}'(a)$. Let $j_L : \mathcal{H}_L \rightarrow \mathcal{H}_+$ be inclusion and $Q_L = j_L j_L^*$ the physical support projection.

Proposition 6 (Support-compression limit). The half-line source derivative

$$\dot{\mathcal{G}}_+ = \left[\left(\frac{d}{d\sigma} \right) (X_+(\sigma)^* X_+(\sigma)) \right]_{\sigma=a} = -\dot{\mathcal{K}}^* \mathcal{K} - \mathcal{K}^* \dot{\mathcal{K}} \quad (72)$$

is trace class. For every $L > 0$,

$$j_L \dot{\mathcal{G}}_L j_L^* = Q_L \dot{\mathcal{G}}_+ Q_L, \quad \|Q_L \dot{\mathcal{G}}_+ Q_L - \dot{\mathcal{G}}_+\|_1 \rightarrow 0 \quad (L \rightarrow \infty). \quad (73)$$

Proof. Apply the full-line output decomposition to sources supported in the positive half-line. Its negative output is $\mathcal{K}(s)$, so (55) proves (72). The identity $\mathcal{L}_L(s) = \mathcal{K}(s)j_L$ gives the exact compression in (73).

For a kernel k , the squared Hilbert–Schmidt norm of its Hankel map on inputs supported after L is $\int_L^\infty (y-L)|k(y)|^2 dy$. The trace-ideal inequality applied to the two products in (72) bounds the error by

$$\begin{aligned} & 2\|\mathcal{K}\|_{\text{HS}} \left(\int_L^\infty (y-L)|k'_a(y)|^2 dy \right)^{\frac{1}{2}} \\ & + 2\|\dot{\mathcal{K}}\|_{\text{HS}} \left(\int_L^\infty (y-L)|k_a(y)|^2 dy \right)^{\frac{1}{2}}. \end{aligned} \quad (74)$$

Both tails tend to zero by weighted square integrability. $\square$

These are trace norms in source coordinates. The physical Weil form is recovered by using the unique sources u satisfying $X_L(a)u = f$ for outputs f in its range. Their norms need not stay bounded as the window grows. Section 6 gives the inverse on this range and an explicit bound. Thus the source limit provides a common operator for the support compressions while leaving the quantitative control of moving physical probes to a further argument.

4. The completed current and its Fourier matrix

The leakage identity expresses the Weil form on completed outputs through the escaped source energy. To compute its arithmetic coefficients, we return to the current: its Hermitian pairing combines the two translation directions. We first write the localized form, then center its smooth prime density against the completion terms, and finally compute every Fourier entry.

4.1. The localized Weil form

For $u, v \in H^1(0, L)$, set

$$q_{u,v}(y) = \langle u, (S_y + S_y^*)v \rangle. \quad (75)$$

Here $q_{u,v}(0) = 2\langle u, v \rangle$ and $q_{u,v}(L) = 0$.

Theorem 7 (Completed current and localized Weil form). For $u, v \in H^1(0, L)$, the form (35) is

$$\begin{aligned} W_L(u, v) = & \int_0^L 2 \cosh\left(\frac{y}{2}\right) q_{u,v}(y) dy - c_L \langle u, v \rangle \\ & - \int_0^L \left(\rho(y) q_{u,v}(y) - \frac{2 \langle u, v \rangle}{e^y - e^{-y}} \right) dy \\ & - \sum_{1 < n < e^L} \frac{\Lambda(n)}{\sqrt{n}} q_{u,v}(\log n). \end{aligned} \quad (76)$$

Under the isometry $(\kappa u)(r) = u(\log(\lambda r))$ from $L^2(0, L)$ to $L^2([\lambda^{-1}, \lambda], d\frac{r}{r})$, this equals the localized Weil form $\mathbf{QW}_\lambda(\kappa u, \kappa v)$ in the normalization of Connes–Consani–Moscovici.

Proof. First take smooth interval vectors. The prime and Γ terms follow from (29) and (23) by taking the two sesquilinear pairings. The kernel of $R_{\frac{1}{2}} + R_{-\frac{1}{2}}$ for $u > t$ is $2 \cosh(\frac{u-t}{2})$. Its Hermitian sum has this kernel on the full square, which gives the first integral in (76). Equivalently the polar operator is

$$2|c_0\rangle\langle c_0| - 2|h_0\rangle\langle h_0|, \quad c_0(t) = \cosh\left(\frac{t - \frac{L}{2}}{2}\right), \quad h_0(t) = \sinh\left(\frac{t - \frac{L}{2}}{2}\right). \quad (77)$$

In this formula $|f\rangle\langle f|$ means the operator $u \mapsto f\langle f, u \rangle$.

For the source comparison, CCM [1, Proposition 3.2, equations (3.14)–(3.18)] expresses the form as its half-distribution applied to the paired correlation. Its polar and prime terms give exactly the corresponding terms of (76). The real-place formula has initial coefficient $(\gamma + \log(4\pi)) \frac{q_{u,v}(0)}{2}$ and an integral over the positive half-line. Since $q_{u,v}$ vanishes beyond L , its tail equals

$$-q_{u,v}(0) \int_L^\infty \frac{1}{e^y - e^{-y}} dy = \left(\frac{q_{u,v}(0)}{2} \right) \log \tanh\left(\frac{L}{2}\right). \quad (78)$$

This gives c_L with the sign in (76). Near zero the relation $q_{u,v}(y) = q_{u,v}(0) + O(y)$ makes the subtracted integrand bounded. At a possible endpoint $n = e^L$, the correlation is zero, so the strict prime cutoff agrees with the source formula. Smooth density in H^1 and the current continuity established after (34) extend the equality to the stated domain. $\square$

On compactly supported smooth functions, centering the interval and converting the pairing convention identifies this form with the differentiated-test-function pairing in [9, Proposition 3.1 and §3.5] and [10, Lemma 3.1]. Theorem 5 obtains the form on completed outputs by varying the arithmetic parameter.

This identity is an identity of forms on the declared vectors. It does not require those vectors to lie in the domain of an unbounded adjoint of J_L . In particular a finite form matrix can be computed from (76) without taking a logarithmic derivative after projection.

4.2. Removing the smooth prime density

The polar term and the smooth part of the prime sum contain the same growing translations. We subtract that smooth density from the prime sum and combine it with the polar terms; the remaining decaying translations then cancel the translation part of the lowest Γ mode. We call this rearrangement centering the completed current. To make the cancellation exact, put

$$a_0 = \psi\left(\frac{1}{4}\right) - \log \pi, \quad \rho_1(q) = \rho(q) - e^{-\frac{q}{2}} = \frac{e^{-5\frac{q}{2}}}{1 - e^{-2q}}, \quad (79)$$

$$R(u) = \sum_{1 < n < u} \frac{\Lambda(n)}{\sqrt{n}} - 2\sqrt{u}, \quad dR(u) = \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} \delta_n(du) - u^{-\frac{1}{2}} du. \quad (80)$$

Here δ_n in the measure denotes the unit atom at n . The measure on $(1, \infty)$ is specified explicitly; its primitive uses the strict sum and has $R(1) = -2$.

Corollary 8 (Centered completed current). Fix $L > 0$. For smooth u, v on $[0, L]$, extend the killed correlation $q_{u,v}(q)$ by zero for $q \geq L$. Then

$$\begin{aligned} W_L(u, v) &= (a_0 + 4)\langle u, v \rangle \\ &\quad + \int_0^\infty \rho_1(q) [2\langle u, v \rangle - q_{u,v}(q)] dq \\ &\quad - \int_{(1, \infty)} q_{u,v}(\log u') dR(u'). \end{aligned} \quad (81)$$

Equivalently, the form is represented on these vectors by the $L^2(0, L)$ -valued expression

$$\begin{aligned} H_L v &= (a_0 + 4)v \\ &\quad + \int_0^\infty \rho_1(q) [2v - S_q v - S_q^* v] dq \\ &\quad - \int_{(1, \infty)} (S_{\log u'} + S_{\log u'}^*) v dR(u'), \end{aligned} \quad (82)$$

so that $W_L(u, v) = \langle u, H_L v \rangle$. The Γ scalar subtraction and its tail extend over the whole positive half-line.

Proof. The subtracted digamma integral [11, equation 5.9.16], after the change of variable $t = 2q$, rewrites the Hermitian Γ part of (18) as

$$a_0 I + \int_0^\infty \rho(q) [2I - S_q - S_q^*] dq. \quad (83)$$

This is an identity on the stated vectors, with the subtraction kept inside the integral. Insert $d\Psi = dR + u^{-\frac{1}{2}} du$ into the prime term of (76). Changing variables $u = e^q$ makes its smooth part $-e^{\frac{q}{2}}(S_q + S_q^*) dq$. The polar term has coefficient $e^{\frac{q}{2}} + e^{-\frac{q}{2}}$. The growing coefficients cancel. Write $\rho = \rho_1 + e^{-\frac{q}{2}}$ in the remaining Γ expression. Its translation part cancels the remaining polar translations, leaving the scalar

$$2 \int_0^\infty e^{-\frac{q}{2}} dq = 4. \quad (84)$$

This proves (81) and (82).

For completeness, zero extension of a smooth interval vector may have endpoint jumps. Nevertheless

$$\|(S_q - I)v\| + \|(S_q^* - I)v\| = O_v(\sqrt{q}) \quad (q \rightarrow 0^+). \quad (85)$$

On the interior this follows from its bounded derivative, and the two boundary strips have length q . Since $\rho_1(q) = O(\frac{1}{q})$ at zero, the vector Γ integral converges there; its tail decays exponentially. The prime measure has only finitely many relevant atoms, and its continuous part is integrated over $1 < u' < e^L$. These estimates justify the vector representing the form. Its integral converges despite the endpoint jumps of the zero extension. $\square$

The lower endpoint can also be checked directly. Write $Q(u) = q_{u_0, v}(\log u)$ for any two smooth interval vectors u_0, v . Then $Q(e^L) = 0$ and integration by parts gives

$$\int_{(1, e^L)} Q(u) dR(u) = 2Q(1) - \int_1^{e^L} R(u) Q'(u) du. \quad (86)$$

The first term is $4\langle u_0, v \rangle$. Thus a primitive formulation absorbs the displayed $+4$ into its lower-end contribution; dropping $R(1) = -2$ would change the current. An atom at $u = e^L$ contributes zero. For $q \geq L$ the shifted terms vanish, whereas $2v \int_L^\infty \rho_1(q) dq$ remains in (82).

The centered formula combines the prime density and polar terms into the signed measure dR and leaves the faster-decaying Γ kernel ρ_1 . No bound for R is needed for this rearrangement. The next calculation uses the measure itself to obtain the Fourier matrix.

4.3. The complete Fourier entries

We now compute the form in the periodic coordinates used by the finite characteristic. Fix an integer $N \geq 1$, and put

$$U_n(t) = L^{-\frac{1}{2}} e^{i\omega_n t}, \quad \omega_n = \frac{2\pi n}{L}, \quad E_N = \text{span}\{U_n : |n| \leq N\}. \quad (87)$$

Let Π_N denote the orthogonal projection onto E_N . The periodic derivative $D_N U_n = \omega_n U_n$ acts on this finite-dimensional space. It is distinct from the absorbing generator A_L of §2. Reflection $\mathcal{R}f(t) = f(L - t)$ sends U_n to U_{-n} , and endpoint evaluation has adjoint

$$\text{ev}_0 f = f(0) = f(L), \quad \delta_N = \text{ev}_0^* = L^{-\frac{1}{2}} \sum_{|n| \leq N} U_n. \quad (88)$$

Thus $\text{ev}_0 f = \langle \delta_N, f \rangle$ and $\|\delta_N\|^2 = \frac{2N+1}{L}$. For $N \geq 1$ it differs from the constant mode U_0 .

The logarithmic coordinate identifies this form with CCM's localized Weil form. For $\lambda = e^{\frac{L}{2}}$, define

$$(\kappa f)(r) = f(\log(\lambda r)), \quad \lambda^{-1} < r < \lambda. \quad (89)$$

Since $d_r^x = dt$, κ is unitary from $\mathcal{H}_L$ to $L^2((\lambda^{-1}, \lambda), d_r^x)$. The multiplicative Fourier modes of CCM are $V_n = \kappa U_n$, and

$$\text{QW}_\lambda^N(\kappa f, \kappa g) = W_L(f, g) = \langle f, H_N g \rangle, \quad f, g \in E_N. \quad (90)$$

Here QW_λ^N denotes the restricted form and H_N its matrix in the orthonormal U coordinates, equivalently the V coordinates after κ ; see [1, Proposition 3.2, equations (3.17)–(3.21), and §5.1]. Thus $(H_N)_{mn} = W_L(U_m, U_n)$. The vector $\kappa \delta_N$ is the endpoint Dirichlet-kernel vector of [1, Corollary 5.6, equation (5.15)]. Reflection commutes with H_N because $\mathcal{R}S_y \mathcal{R} = S_y^*$. The even and odd dimensions are $N + 1$ and N .

Put $\lambda_k = 2k + \frac{1}{2}$. The original Γ multiplier, including its lowest mode, is

$$a_\Gamma(\xi) = \Re \psi\left(\frac{1}{4} + i\frac{\xi}{2}\right) - \log \pi = a_0 + 2 \sum_{k \geq 0} \frac{\xi^2}{\lambda_k(\lambda_k^2 + \xi^2)}. \quad (91)$$

For real ξ define the real odd function

$$\begin{aligned} \mathfrak{a}_x(\xi) = & - \sum_{1 < n < x} \frac{\Lambda(n)}{\sqrt{n}} \sin\left(\xi \log\left(\frac{x}{n}\right)\right) \\ & + \frac{(\sqrt{x}-1)\xi}{\xi^2 + \frac{1}{4}} + \xi \sum_{k \geq 1} \frac{1 - x^{-\lambda_k}}{\lambda_k^2 + \xi^2}. \end{aligned} \quad (92)$$

We call $\mathfrak{a}_x$ the arithmetic interpolation. For a differentiable function f , write

$$f[u, v] = \begin{cases} \frac{f(u)-f(v)}{u-v} & u \neq v \\ f'(u) & u = v \end{cases}. \quad (93)$$

The Γ series in (92) and its derivatives converge locally uniformly on the real line: on each compact set their tails are bounded by constant multiples of λ_k^{-2} . In particular the confluent values are defined at every real frequency, including zero.

Theorem 9 (The complete arithmetic Fourier matrix). For every $L > 0$ and integer $N \geq 1$, the matrix of (76) in the basis $U_j(t) = L^{-\frac{1}{2}} e^{i\omega_j t}$, $|j| \leq N$, is

$$(H_N)_{jk} = a_\Gamma(\omega_j) \delta_{jk} + \frac{2}{L} \mathfrak{a}_x[\omega_j, \omega_k]. \quad (94)$$

Here δ_{jk} is the Kronecker symbol. This includes the genuine diagonal derivative of (92). No hypothesis on the least eigenvalue is required.

The off-diagonal entries determine displacement, but they alone do not determine the ground. The proof calculates the diagonal separately and then shows why the same interpolation gives its confluent value. This realizes the divided-difference structure of CCM [1, Lemma 5.1] and Connes–van Suijlekom [6, Proposition 4.1] for the completed arithmetic form.

Proof. For $0 \leq q \leq L$, direct integration over the overlap of the shifted intervals gives

$$\langle U_j, (S_q + S_q^*) U_k \rangle = \begin{cases} -\frac{2}{L} \frac{\sin(\omega_j q) - \sin(\omega_k q)}{\omega_j - \omega_k} & j \neq k \\ 2(1 - \frac{q}{L}) \cos(\omega_j q) & j = k \end{cases}. \quad (95)$$

For $q \geq L$ the correlation is zero. Define an auxiliary interpolation by the actual finite integrals,

$$A(\xi) = \int_0^L \rho_1(q) \sin(\xi q) dq + \int_{(1,x)} \sin(\xi \log u) dR(u). \quad (96)$$

Inserting (95) into (81) proves that the off-diagonal entry is $(\frac{2}{L})A[\omega_j, \omega_k]$. At a Fourier frequency, elementary integration gives $A(\omega_j) = \mathfrak{a}_x(\omega_j)$; the three component formulas are supplied in Appendix C. This proves the off-diagonal identity. The lattice equality alone does not determine either derivative.

For the diagonal put $\omega = \omega_j$. The multiplier of the first two terms of (82) on an uncut exponential is

$$a_1(\omega) = a_0 + 4 + 2 \int_0^\infty \rho_1(q) (1 - \cos(\omega q)) dq = a_\Gamma(\omega) + \frac{1}{\omega^2 + \frac{1}{4}}. \quad (97)$$

Using the diagonal line of (95) instead of differentiating the lattice identity gives

$$(H_N)_{jj} = a_1(\omega) + \frac{2}{L}A'(\omega) + 2 \int_L^\infty \rho_1(q) \cos(\omega q) dq - 2 \int_{(1,x)}^\infty \cos(\omega \log u) dR(u). \quad (98)$$

The scalar Γ tail in this equation is part of the original form.

The component calculations in Appendix C give the exact combined repair

$$\left(\frac{2}{L}\right)(\mathbf{a}'_x(\omega) - A'(\omega)) = \frac{1}{\omega^2 + \frac{1}{4}} + 2 \int_L^\infty \rho_1(q) \cos(\omega q) dq - 2 \int_{(1,x)}^\infty \cos(\omega \log u) dR(u). \quad (99)$$

Substitution in (98) proves the diagonal in (94), including zero frequency. The tail beyond L is essential to this equality. $\square$

The oddness of $\mathbf{a}_x$ and evenness of a_Γ show that the matrix is real symmetric and satisfies $(H_N)_{-j,-k} = (H_N)_{jk}$. When x crosses a prime power, the strict sum specifies the entry exactly; at $x = n$ its newly arriving value and diagonal derivative contribution vanish because $\log(\frac{x}{n}) = 0$.

To compare centered coordinates, let $r = t - \frac{L}{2}$ and $e_j(r) = L^{-\frac{1}{2}}e^{i\omega_j r}$. The unitary translation sends U_j to $(-1)^j e_j$. Hence

$$(H_N^{\text{cent}})_{jk} = (-1)^{j+k} (H_N)_{jk} = a_\Gamma(\omega_j) \delta_{jk} + \frac{2(-1)^{j+k}}{L} \mathbf{a}_x[\omega_j, \omega_k]. \quad (100)$$

Endpoint evaluation in the centered basis is $e_j(-\frac{L}{2}) = \frac{(-1)^j}{\sqrt{L}}$; in the manuscript basis it is $U_j(0) = \frac{1}{\sqrt{L}}$. The physical derivative remains diagonal with eigenvalues ω_j in both bases. Thus neither the endpoint row nor the generator acquires an additional normalization factor.

In the integer-frequency convention of CCM, (94) gives

$$\beta_j = \frac{\mathbf{a}_x(\omega_j)}{\pi}, \quad (H_N)_{jk} = \frac{\beta_j - \beta_k}{j - k} \quad (j \neq k). \quad (101)$$

This also gives a precise comparison with the distributional interpolation of [6, Proposition 4.1]. Denote that interpolation by ψ_{CvS} , reserving ψ for the digamma function. For the present distribution its integer values and derivatives are

$$\psi_{\text{CvS}}(j) = \beta_j, \quad \psi'_{\text{CvS}}(j) = a_\Gamma(\omega_j) + \frac{2}{L} \mathbf{a}'_x(\omega_j). \quad (102)$$

The second equality retains the Γ diagonal. Thus ψ_{CvS} differs from $u \mapsto \pi^{-1} \mathbf{a}_x(\frac{2\pi u}{L})$.

The off-diagonal formula gives

$$[D_N, H_N] = b_N \text{ev}_0 - \delta_N b_N^*, \quad b_N = \frac{2\pi}{\sqrt{L}} \sum_{|j| \leq N} \beta_j U_j. \quad (103)$$

Indeed both off-diagonal entries are $(\frac{2\pi}{L})(\beta_j - \beta_k)$, and both diagonals vanish. The vector b_N is odd and δ_N is even. The off-diagonal calculation supplies displacement; the independent diagonal calculation completes the arithmetic matrix on which selection acts.

5. Boundary selection and the finite characteristic

We now apply the finite mechanism of CCM [1, §5] and Connes–van Suijlekom [6, §5] to this matrix. Under the hypotheses below, displacement forces the least eigenline to have nonzero endpoint evaluation. Subtracting its eigenvalue and removing that line gives a positive inner product. A rank-one correction then makes the induced derivative self-adjoint. The determinant ratio of the corrected derivative on the full section, multiplied by the free sinc factor, gives the centered Fourier transform of the endpoint-normalized ground divided by L .

In the continuum, Suzuki constructs real-zero characteristics from self-adjoint derivative extensions in a Weil metric shifted strictly below the spectral bottom [10, Lemma 6.2 and Theorem 1.5]. Here we use the CCM characteristic of the selected finite ground, subtracting its actual least eigenvalue and passing to the quotient by the resulting null line.

Throughout this section suppose that H_N is real symmetric, commutes with reflection, satisfies (103) with odd b_N , and has a simple least eigenvalue with an even eigenvector. The arithmetic Weil matrix is one such matrix whenever this last spectral hypothesis holds. Let ε_N be its full least eigenvalue, $P_{\min,N}$ the ground projector, and $T_N = H_N - \varepsilon_N I$. Then $T_N \geq 0$ and $\ker T_N = \text{Ran } P_{\min,N}$ is a single even line. We refer to this spectral condition as the full simple-even hypothesis. We call nonvanishing of the ground line under endpoint evaluation boundary brightness.

Lemma 10 (Boundary brightness). Under these hypotheses, $q_N = \langle \delta_N, P_{\min,N} \delta_N \rangle > 0$. The vector

$$\xi_N = \frac{P_{\min,N} \delta_N}{q_N} \quad (104)$$

is even, lies in the ground line and satisfies $\text{ev}_0 \xi_N = 1$ and $\|\xi_N\|^2 = \frac{1}{q_N}$.

Proof. If an even ground vector u satisfied $\text{ev}_0 u = 0$, then $b_N^* u = 0$ by parity. The identity (103) would give $T_N D_N u = 0$. But $D_N u$ is odd, whereas the kernel is even; thus $D_N u = 0$. The simple zero eigenspace of D_N consists of constants, so $u = 0$, a contradiction. This proves brightness of the ground line and hence $q_N > 0$. The remaining identities follow from $P_{\min,N}^2 = P_{\min,N} = P_{\min,N}^*$. $\square$

The normalization matches CCM’s physical endpoint vector. In the abstract integer-coordinate lemmas of [1, Lemma 5.4] and [6, Lemma 5.2], the corresponding vectors and derivative are

$$\eta = \sqrt{L} \delta_N, \quad \xi_{\text{abs}} = \frac{\xi_N}{\sqrt{L}}, \quad D_{\text{int}} = \frac{L}{2\pi} D_N. \quad (105)$$

Indeed $\langle \eta, \xi_{\text{abs}} \rangle = 1$. Consequently

$$\frac{2\pi}{L} [D_{\text{int}} f - (D_{\text{int}} \xi_{\text{abs}}) \langle \eta, f \rangle] = D_N f - (D_N \xi_N) \text{ev}_0 f, \quad f \in E_N. \quad (106)$$

The correction below is already in physical frequencies, as in [1, Theorem 5.10, equation (5.26)].

The shifted form vanishes precisely on the ground line, so it becomes a positive inner product after that line is removed. To act on the quotient, the derivative must preserve the ground line. The following boundary correction makes it vanish there.

Proposition 11 (The corrected derivative). Set

$$D'_N = D_N - (D_N \xi_N) \text{ev}_0. \quad (107)$$

Then $D'_N \xi_N = 0$ and $T_N D'_N = D_N'^* T_N$. The induced operator D''_N on $\frac{E_N}{\mathbb{C}\xi_N}$ is self-adjoint for the positive inner product $\langle u, T_N w \rangle$ on equivalence classes. In particular all eigenvalues of D'_N and D''_N are real.

Proof. Applying (103) to ξ_N gives $T_N D_N \xi_N = -b_N$, since $\text{ev}_0 \xi_N = 1$ and $b_N^* \xi_N = 0$. Hence

$$T_N D'_N = T_N D_N + b_N \text{ev}_0 = D_N T_N + \delta_N b_N^* = D_N'^* T_N. \quad (108)$$

The form descends to the quotient because its kernel is exactly $\mathbb{C}\xi_N$, and is positive definite there. The induced operator is therefore self-adjoint in a finite-dimensional Hilbert space. Finally the full characteristic polynomial factors as $(-z) \det(D''_N - zI)$, since D'_N kills its invariant ground line. This also accounts for the additional zero eigenvalue of D'_N . $\square$

The proof uses positivity after subtracting the actual least eigenvalue. It is the finite mechanism underlying CCM's perturbed scaling operator and the real-zero theorem of [6, Theorem 5.6]. No positivity of the unshifted Weil form enters it.

5.1. A determinant observed at the endpoint

For z off the finite free spectrum define

$$\mathcal{T}_N(z) = \frac{\det(D'_N - zI)}{\det(D_N - zI)}, \quad \Theta_N(z) = \text{sinc}\left(\frac{Lz}{2}\right) \mathcal{T}_N(z), \quad (109)$$

where $\text{sinc}(w) = \frac{\sin(w)}{w}$ with value 1 at zero. Define independently the centered Fourier transform

$$\hat{\xi}_N(z) = \int_0^L \xi_N(t) e^{-iz(t-\frac{L}{2})} dt. \quad (110)$$

Since $t - \frac{L}{2} = \log r$ under κ , this is also the multiplicative Fourier transform of $\kappa \xi_N$ used in [1, Proposition 5.9].

Theorem 12 (Finite boundary characteristic). With the standing full simple-even ground hypotheses,

$$\mathcal{T}_N(z) = -z \text{ev}_0 (D_N - zI)^{-1} \xi_N, \quad \Theta_N(z) = \frac{\hat{\xi}_N(z)}{L}. \quad (111)$$

The second expression extends Θ_N to an entire function with only real zeros. If $\xi_N = \sum \xi_{N,n} U_n$, its free-lattice values are

$$\Theta_N(\omega_j) = \begin{cases} (-1)^j \frac{\xi_{N,j}}{\sqrt{L}} & |j| \leq N \\ 0 & |j| > N \end{cases}. \quad (112)$$

Proof. The finite determinant lemma applied to (107) gives

$$\mathcal{T}_N(z) = 1 - \text{ev}_0 (D_N - zI)^{-1} D_N \xi_N = -z \text{ev}_0 (D_N - zI)^{-1} \xi_N, \quad (113)$$

using $\text{ev}_0 \xi_N = 1$. Direct integration of one Fourier vector gives

$$\int_0^L U_n(t) e^{-iz(t-\frac{L}{2})} dt = \frac{2 \sin(\frac{Lz}{2})}{\sqrt{L}(z - \omega_n)}. \quad (114)$$

Summing proves

$$\hat{\xi}_N(z) = -2 \sin\left(\frac{Lz}{2}\right) \text{ev}_0(D_N - zI)^{-1} \xi_N = L\Theta_N(z). \quad (115)$$

This is an entire function, and evaluation of the removable singularities gives (112), including $j = 0$. Alternatively (112) follows by Fourier orthogonality. For nonreal z , both $\sin(\frac{Lz}{2})$ and $\det(D_N - zI)$ are nonzero. All eigenvalues of D'_N are real by the preceding proposition, so (111) cannot vanish there. The transform is not identically zero because ξ_N is nonzero; this also follows from its finite Fourier coefficients in (112). $\square$

The characteristic depends on the ground line's position relative to the derivative and the endpoint row. The projector formula removes an eigenvector phase, but it still uses the spectral selection. For instance, finite spectral calculus also gives

$$\xi_N = \frac{\text{adj}(H_N - \varepsilon_N I) \delta_N}{\text{ev}_0 \text{adj}(H_N - \varepsilon_N I) \delta_N}. \quad (116)$$

To verify this, diagonalize H_N : its adjugate at ε_N is the ground projector multiplied by the product of all positive gaps. The denominator is positive by brightness. The energy ε_N remains an input to this formula.

6. Range, projection and moving limits

The preceding results place all physical support windows inside one family of bounded half-line operators and their source-Gram derivatives inside one trace-class operator. They control the forward operators and their derivatives in common source coordinates. Transporting changing periodic spectral minimizers through this source-coordinate representation requires estimates for their preimages under the projected completed operators. The unprojected range is fixed by an absorbing boundary condition, which differs from the periodic condition used for the ground construction.

For $f \in \text{Ran } X_L(a)$, the unique source u satisfying $X_L(a)u = f$ can be written explicitly. This makes its possible growth visible before considering moving periodic sections.

Resolvent algebra on $\text{Dom } A_L$ gives

$$U_L f := X_L(a)^{-1} f = \frac{1}{2\pi} Z_L(a)^{-1} \left[\left(\frac{9}{2} - A_L \right) f + 6R_{-\frac{1}{2}} f \right], \quad f \in \text{Dom } A_L. \quad (117)$$

The inverse acts on the terminal-zero domain with its derivative norm.

Combining the Möbius factor with the growing Volterra resolvent gives the elementary bound

$$\|U_L f\|_2 \leq C e^{\frac{L}{2}} \|f\|_{H^1(0,L)}, \quad f \in \text{Dom } A_L, \quad (118)$$

with an absolute constant; Appendix B supplies the calculation. This upper bound deteriorates with L and supplies no uniform control on changing physical outputs without further estimates. The uniform trace bound in (73) concerns source coordinates. Applying it to changing physical vectors also requires control of their preimages.

Proposition 13 (The completed range and the endpoint hyperplane). For fixed $L > 0$ and $N \geq 1$, assume that the least eigenvalue of the full matrix H_N is simple and has an even eigenvector. Then

$$\text{Ran } X_L \left(\frac{1}{2} \right) \cap E_N = \ker \text{ev}_0. \quad (119)$$

Let $g > 0$ be the first positive eigenvalue of T_N , and let

$$\mu_N = \min_{0 \neq u \in \ker \text{ev}_0} \frac{\langle u, H_N u \rangle}{\|u\|^2}. \quad (120)$$

Then

$$\mu_N - \varepsilon_N \geq \frac{gLq_N}{2N+1} > 0. \quad (121)$$

Proof. The range in (28) consists exactly of terminal-zero H^1 functions. For a periodic Fourier polynomial, the terminal value equals the initial value. This gives (119) and excludes ξ_N , whose endpoint value is 1.

Let f_0 be a unit ground vector. The spectral gap gives $T_N \geq g(I - P_{\min, N})$. Among unit vectors u orthogonal to δ_N , the largest possible value of $|\langle f_0, u \rangle|^2$ is

$$1 - \frac{|\langle \delta_N, f_0 \rangle|^2}{\|\delta_N\|^2} = 1 - \frac{q_N}{\|\delta_N\|^2}. \quad (122)$$

Therefore $\langle u, T_N u \rangle \geq g \frac{q_N}{\|\delta_N\|^2}$ on that unit sphere. Since $\|\delta_N\|^2 = \frac{2N+1}{L}$, (121) follows. The minimum exists because this is a nonzero finite-dimensional subspace. $\square$

Remark (Projected range). For every fixed $L > 0$ and $N \geq 1$,

$$\Pi_N \text{Ran } X_L \left(\frac{1}{2} \right) = E_N. \quad (123)$$

Indeed (28) identifies the range with $\text{Dom } A_L$, which contains $C_c^\infty(0, L)$ and is dense in $\mathcal{H}_L$. Its orthogonal projection is dense in E_N and is a linear subspace. Finite dimensionality makes that image closed, proving (123). No ground hypothesis is needed for this identity.

The nonzero endpoint value excludes ξ_N from $\text{Ran } X_L \left(\frac{1}{2} \right)$, whereas every finite Fourier vector has a preimage under $\Pi_N X_L \left(\frac{1}{2} \right)$. Thus representability after projection is automatic at each fixed section. For the continuum localized form at a fixed interval, Suzuki proves form-norm approximation of every minimizer by compactly supported smooth functions [10, Corollary 1.2]. The estimates sought here concern the changing finite ground states and their preimages under the projected completed maps as L and N grow.

There are two further elementary restrictions on transporting the construction. First, physical support inclusion intertwines the directed shifts exactly, but it does not preserve periodic finite sections. The zero extension of a boundary-bright periodic function on $(0, L_1)$ has a jump at L_1 inside a larger interval. It is not in $H^1(0, L_2)$ and hence is not a Fourier polynomial on that interval. Second, for an orthogonal projection Π the exact identity is

$$(\Pi S_a \Pi)(\Pi S_b \Pi) - \Pi S_{a+b} \Pi = -\Pi S_a (I - \Pi) S_b \Pi. \quad (124)$$

This follows by inserting $I = \Pi + (I - \Pi)$ between the shifts. Form restriction remains valid, whereas products of projected factors contain the displayed defect.

We now restore the support dependence by writing $\hat{\xi}_{L,N}$ for the transform $\hat{\xi}_N$ of §5. At every pair satisfying the full simple-even hypothesis, the finite real-zero theorem makes the following normalization well defined:

$$\mathcal{F}_{L,N}(z) = \frac{\hat{\xi}_{L,N}(z)}{\hat{\xi}_{L,N}(z_*)}, \quad z_* = \frac{i}{4}. \quad (125)$$

For the target, put

$$\Xi(z) = \xi\left(\frac{1}{2} + iz\right), \quad \xi(s) = \frac{1}{2}s(s-1)\Gamma_R(s)\zeta(s). \quad (126)$$

The target denominator is nonzero: $\Xi(\frac{i}{4}) = \xi(\frac{1}{4})$, and the alternating Dirichlet series shows that $\zeta(s)$ has no real zero for $0 < s < 1$. Grouping consecutive terms gives a strictly positive alternating numerator and $1 - 2^{1-s} < 0$.

We call an entire limit G zero-retaining if every zero of Ξ is also a zero of G . The following criterion isolates the moving conclusion needed from the finite real-zero construction.

Proposition 14 (Zero-retaining entire-limit criterion). Let (L_j, N_j) be one sequence with $L_j \rightarrow \infty$ and $N_j \rightarrow \infty$, satisfying the full simple-even hypothesis at every pair. Suppose $\mathcal{F}_{L_j, N_j}$ converges locally uniformly on all of $\mathbb{C}$ to an entire function G . If every zero of Ξ is a zero of G , then the Riemann hypothesis holds.

Proof. The common normalization gives $G(\frac{i}{4}) = 1$, so G is not identically zero. Every approximant has only real zeros by the finite characteristic theorem. Hurwitz's theorem on each open half-plane therefore makes G zero-free there: an identically zero restriction would make the entire function zero everywhere. The assumed inclusion of zero sets then excludes nonreal zeros of Ξ , which is the Riemann hypothesis. $\square$

The hypothesis asks for one sequence satisfying the stated spectral conditions and one nonzero entire limit retaining every Ξ zero. A factorization $G = A\Xi$ with A entire would suffice, but also requires retention of multiplicities, which zero-set inclusion alone does not assert. A quotient on a zero-free disk supplies no such global retention. The proposition uses whole-plane convergence; a disk limit would require a separate upgrade argument.

A distinguished stronger target is to prescribe, from the arithmetic data, pairs (L_j, N_j) with $L_j \rightarrow \infty$ and integer $N_j \rightarrow \infty$, for which the full least eigenvalue is simple with an even eigenvector at every pair, and such that

$$\sup_{z \in K} \left| \mathcal{F}_{L_j, N_j}(z) - \frac{\Xi(z)}{\Xi(z_*)} \right| \rightarrow 0 \quad \text{for every compact } K \subset \mathbb{C}. \quad (127)$$

This is local uniform convergence on all of $\mathbb{C}$ along that prescribed sequence, not an assertion about every growing choice of cutoffs.

The finite companion [4, Theorem 5] supplies compactness and zero retention under explicit moving hypotheses. On one unbounded integer sequence x_j , put $L_j = \log x_j$ and let $N_j \geq 1$ be arbitrary integers. Assume that the full least eigenvalue is simple with an even eigenvector at each pair and that the complete characteristic means are uniformly bounded. After centering the interval, its anchor-normalized characteristic is exactly (125). The complete mean gives local boundedness and forces N_j to grow at least quadratically in L_j ; the anchor normalization makes every entire cluster nonzero. Gaussian tests then give the companion's current passage and retention of every zero of Ξ in each cluster. Its constant-mode transfer controls the complete filtered shifted energy without a uniform positive excitation gap.

Existence of such an unbounded selected family and its uniform mean bound remain unresolved and must hold on the same sequence. Exact identification in (127) is a stronger target. CCM [1, §§7–8] separates fixed-window ground approximation from this moving-support question. The common half-line family supplies the forward source-coordinate framework, while the companion’s finite argument supplies another sufficient route to Proposition 14. It does not establish uniform estimates for projected-source preimages; those remain a separate transport problem.

Appendix A. The regularized determinant and its phase

For completeness we record the exact relation to the spectral determinant in CCM. This also fixes the normalization needed to compare moving characteristics. At a fixed pair (L, N) , adjoin the unchanged free Fourier complement to D_N'' , using the positive shifted metric on the finite quotient and the ordinary metric on the complement. Let $\Delta_{L,N}(z)$ be its ζ -regularized determinant, with the source cut

$$(-1)^{-w} = e^{-i\pi w}. \quad (128)$$

CCM [1, Lemma 5.8] computes the free periodic determinant as $1 - e^{-iLz}$. Finite replacement of the free sector, and removal of the ground null line, give

$$\Delta_{L,N}(z) = (1 - e^{-iLz}) \frac{\det(D_N'' - zI)}{\det(D_N - zI)}. \quad (129)$$

Since $\det(D_N' - zI) = (-z) \det(D_N'' - zI)$, (111) implies

$$\Delta_{L,N}(z) = (1 - e^{-iLz}) \operatorname{ev}_0(D_N - zI)^{-1} \xi_N = -ie^{\frac{-iLz}{2}} \hat{\xi}_{L,N}(z). \quad (130)$$

The equalities initially hold away from the free lattice and extend through it. This is the normalization of CCM’s Theorem 5.10. Retaining the ground null line instead multiplies (130) by $-z$.

At the nonreal normalization point z_* , the normalized Fourier observation therefore satisfies

$$\frac{\hat{\xi}_{L,N}(z)}{\hat{\xi}_{L,N}(z_*)} = e^{\frac{iL(z-z_*)}{2}} \frac{\Delta_{L,N}(z)}{\Delta_{L,N}(z_*)}. \quad (131)$$

Thus the determinant formulation of (127) compares the ratios on the right of (131), including the exponential, uniformly on every compact subset of $\mathbb{C}$ along the same pairs (L_j, N_j) . The exponential depends on L and has varying modulus off the real axis; convergence of the ungauged determinant ratios is a different condition.

Appendix B. Parameter estimates and the source inverse

This appendix proves the parameter-completion theorem and the multiplier formula (53). The two polar factors produce a mean-zero seed. Comparing its arithmetic samples with its zero integral then gives the weighted kernel estimates needed for both the forward convolution and the leakage operator. The final calculation proves the inverse on $\operatorname{Ran} X_L(a)$ and its bound used in §6.

The seed calculation. Write a causal kernel as $e^{-sy}h(e^{-y})$. Convolution by e^{-by} acts on the seed by

$$T_{b-s}h(u) = u^{b-s} \int_u^1 t^{-(b-s)-1} h(t) dt. \quad (132)$$

The Γ seed is $c_\alpha(1 - u^2)^{\alpha-1}$. Since $\frac{5}{2} - s = 2\alpha$, the first polar factor is $I - 2\alpha T_{2\alpha}$. The identity

$$\left(\frac{d}{du}\right) [u^{-2\alpha}(1 - u^2)^\alpha] = -2\alpha u^{-2\alpha-1}(1 - u^2)^{\alpha-1} \quad (133)$$

shows that this factor leaves $c_\alpha u^2(1 - u^2)^{\alpha-1}$. Applying $I - 2\alpha T_{2\alpha-1}$ gives (51).

Its integral is zero. Indeed absolute Fubini is valid for $\Re\alpha > 0$: putting $A = \Re\alpha$, the absolute double integral is

$$\int_0^1 u^{2A-1} \int_u^1 t^{2-2A}(1 - t^2)^{A-1} dt du = \frac{1}{2A} \int_0^1 t^2(1 - t^2)^{A-1} dt < \infty. \quad (134)$$

For the complex powers, the inner integral after reversing order is $\frac{t^{2\alpha}}{2\alpha}$. It cancels exactly the integral of the first term in (51). When $\alpha = 1$, direct integration gives $h_1(u) = 2\pi(3u^2 - 2u)$.

Uniform endpoint estimates. On the closed disk $|s - a| \leq \frac{1}{4}$ we have $\frac{7}{8} \leq \Re\alpha \leq \frac{9}{8}$. The Γ prefactor is uniformly bounded and the integral in (51) is uniformly bounded. Consequently

$$|h_\alpha(u)| \leq C(1 - u)^{-\frac{1}{8}}. \quad (135)$$

Differentiating before taking absolute values gives

$$\begin{aligned} \frac{h'_\alpha(u)}{c_\alpha} &= (2 + 2\alpha)u(1 - u^2)^{\alpha-1} - 2(\alpha - 1)u^3(1 - u^2)^{\alpha-2} \\ &\quad - 2\alpha(2\alpha - 1)u^{2\alpha-2} \int_u^1 t^{2-2\alpha}(1 - t^2)^{\alpha-1} dt. \end{aligned} \quad (136)$$

Hence

$$|h'_\alpha(u)| \leq C[u^{-\frac{1}{4}} + (1 - u)^{-\frac{9}{8}}]. \quad (137)$$

The same estimates give $h_\alpha(0) = 0$ by continuity. For each fixed $0 < u < 1$, dominated integration shows that $h_\alpha(u)$ is holomorphic in s on a neighborhood of the closed disk. No value at $u = 1$ is needed.

Arithmetic cancellation with threshold errors. Let $A > 1$ be nonintegral, $m = \lfloor A \rfloor$ and $\theta = A - m$. Comparing the first $m - 1$ samples with the integrals over the corresponding cells of length $\frac{1}{A}$ gives

$$\left| \sum_{n=1}^{m-1} h_\alpha\left(\frac{n}{A}\right) - A \int_0^{\frac{m-1}{A}} h_\alpha(u) du \right| \leq \int_0^{\frac{m-1}{A}} |h'_\alpha(u)| du \leq CA^{\frac{1}{8}}. \quad (138)$$

The final sample is bounded by $CA^{\frac{1}{8}}\theta^{-\frac{1}{8}}$, while

$$A \int_{\frac{m-1}{A}}^1 |h_\alpha(u)| du \leq CA^{\frac{1}{8}}. \quad (139)$$

For $m = 1$ the first sum and integral are empty and these bounds still hold. Using the zero integral of the seed proves

$$\left| \sum_{1 \leq n < A} h_\alpha\left(\frac{n}{A}\right) \right| \leq CA^{\frac{1}{8}}(1 + \theta^{-\frac{1}{8}}). \quad (140)$$

This estimate retains the singularity near each threshold, rather than replacing it by a false uniform pointwise bound.

With $A = e^y$ and $\Re s \geq \frac{1}{4}$, (140) gives the common majorant

$$|k_s(\log A)| \leq CA^{-\frac{1}{8}}(1 + \{A\}^{-\frac{1}{8}}) =: M(\log A), \quad (141)$$

away from the integer thresholds. Set $M = 0$ at those thresholds. Split the A integral into $(m, m+1)$ and use $dy = d\frac{A}{A}$. The required integrals are bounded by constant multiples of

$$\sum_{m \geq 1} m^{-\frac{9}{8}} \int_0^1 (1 + \theta^{-\frac{1}{8}}) d\theta, \quad (142)$$

$$\sum_{m \geq 1} (1 + \log(m+1)) m^{-\frac{5}{4}} \int_0^1 (1 + \theta^{-\frac{1}{8}})^2 d\theta. \quad (143)$$

Both converge. Therefore $M \in \mathcal{B}$, uniformly for the closed disk of parameters.

For almost every y , (52) is a finite holomorphic sum. Dominated convergence first gives continuity in $\mathcal{B}$. Apply the scalar Cauchy formula on a parameter circle within D , and use the same majorant to interchange its contour integral with the kernel. The resulting Cauchy formula holds in the Banach space $\mathcal{B}$; its difference quotients prove holomorphy, and its Cauchy estimates bound every fixed derivative on smaller closed disks. This proves the kernel assertions of the theorem. The maps from L^1 kernels to the two bounded convolution operators have norm at most one, so their operator-norm holomorphy follows as well.

Finite restriction. On a source supported in $(0, L)$, the Euler sum terminates. A term indexed by n in the completed product contributes, after the shift of the convolution variable,

$$n^{-s} e^{-s(y - \log n)} h_\alpha(e^{-(y - \log n)}) = e^{-sy} h_\alpha(ne^{-y}), \quad y > \log n. \quad (144)$$

Finite Fubini gives (52). For the positive-half-line output only $y < L$ can contribute, so this product restricts exactly to $X_L(s)$ and preserves $\mathcal{H}_L$. At $n = e^L$ the shift is killed; at $y = \log n$ the representative of the kernel is immaterial. The union of these support subspaces is dense, so this determines the bounded extension uniquely. At a the preceding seed calculation identifies it with the critical operator of Theorem 1.

The full-line multiplier. For sufficiently large $\Re w$, all arithmetic sums and β integrals converge absolutely. The prime, Γ and polar factors respectively give

$$\zeta(s+w), \quad \frac{\Gamma_R(s+w)}{\Gamma_R(\frac{5}{2}+w)}, \quad \frac{(s+w)(s+w-1)}{(\frac{5}{2}+w)(\frac{3}{2}+w)}. \quad (145)$$

Their product is (53), using the conventional normalization of ξ in [11, equation 25.4.4]. Both sides are holomorphic for $\Re w > 0$. On the integral side this follows from $k_s \in L^1$; on the displayed quotient side the reference factors have no zero or pole there. Analytic continuation therefore gives equality throughout that half-plane. Letting $w = \varepsilon - i\omega$ with ε decreasing to zero is justified by dominated convergence against $|k_s|$. Backward convolution has precisely this Fourier multiplier. Finally [11, equation 25.4.3] and real type of ξ prove the real-parameter symmetry used in (55). The fixed denominator is the reference Γ and polar factor; there is no reference ζ factor.

The inverse on the completed range. On $\text{Dom } A_L$, the following calculation proves (117) and (118).

Indeed $X_L(a) = 2\pi R_{\frac{3}{2}}(I - 2R_{\frac{3}{2}})Z_L(a)$, while $(I - 2R_{\frac{3}{2}})^{-1} = I + 2R_{-\frac{1}{2}}$ and $R_{-\frac{1}{2}}A_L = -\frac{1}{2}R_{-\frac{1}{2}} - I$ on this domain. Multiplication gives (117).

The useful elementary estimate combines the Möbius factor with the growing resolvent before taking norms. Put $b_\mu(A) = \sum_{n \leq A} \frac{\mu(n)}{n}$, for $A \geq 1$.

The divisor identity

$$\sum_{n \leq A} \mu(n) \left\lfloor \frac{A}{n} \right\rfloor = 1 \quad (146)$$

implies $Ab_\mu(A) = 1 + \sum_{n \leq A} \mu(n) \left\{ \frac{A}{n} \right\}$, hence $|b_\mu(A)| \leq 2$. Inserting the finite Möbius inverse in the resolvent integral gives, for almost every $0 < t < L$,

$$(Z_L(a)^{-1}R_{-\frac{1}{2}}f)(t) = \int_t^L e^{\frac{w-t}{2}} b_\mu(e^{w-t}) f(w) dw. \quad (147)$$

The additional factor $n^{-\frac{1}{2}}$ from the shifted exponential is what changes the coefficient to $\frac{\mu(n)}{n}$. Young's inequality and $\|Z_L(a)^{-1}\| \leq \sum_{n < e^L} n^{-\frac{1}{2}} \leq Ce^{\frac{L}{2}}$ give (118), with an absolute constant.

Appendix C. The true diagonal of the Fourier matrix

The auxiliary interpolation A in (96) is built from the actual killed integrals. Its values agree with $\mathbf{a}_x$ at the Fourier lattice, but their derivatives differ. We calculate those differences before taking any confluent value.

Split A into its prime, continuous-density and Γ terms, denoted A_p, A_c, A_g . Elementary integration gives

$$A_p(\xi) = \sum_{1 < n < x} \frac{\Lambda(n)}{\sqrt{n}} \sin(\xi \log n), \quad (148)$$

$$A_c(\xi) = -\frac{\sqrt{x} \left[\frac{1}{2} \sin(\xi L) - \xi \cos(\xi L) \right] + \xi}{\xi^2 + \frac{1}{4}}, \quad (149)$$

$$A_g(\xi) = \sum_{k \geq 1} \frac{\xi - x^{-\lambda_k} [\lambda_k \sin(\xi L) + \xi \cos(\xi L)]}{\lambda_k^2 + \xi^2}. \quad (150)$$

At $\xi = \omega_j$, the identities $\sin(\xi L) = 0$ and $\cos(\xi L) = 1$ give $A(\omega_j) = \mathbf{a}_x(\omega_j)$. These are the lattice values used in §4.

Put $\omega = \omega_j$. Write $\mathbf{a}_x = \mathbf{a}_p + \mathbf{a}_c + \mathbf{a}_g$ according to (92). Differentiating the three displayed formulas for A_p, A_c, A_g gives the corrections

$$\begin{aligned} \mathbf{a}'_p(\omega) - A'_p(\omega) &= -L \sum_{1 < n < x} \frac{\Lambda(n)}{\sqrt{n}} \cos(\omega \log n), \\ \mathbf{a}'_c(\omega) - A'_c(\omega) &= \frac{L\sqrt{x}}{2(\omega^2 + \frac{1}{4})}, \\ \mathbf{a}'_g(\omega) - A'_g(\omega) &= L \sum_{k \geq 1} \frac{\lambda_k x^{-\lambda_k}}{\lambda_k^2 + \omega^2}. \end{aligned} \quad (151)$$

For the prime line, differentiate $-\sin(\xi(L - \log n))$ before putting $\xi = \omega$. For the other lines differentiate the displayed formulas for A_c and A_g before using the Fourier lattice relations. Termwise differentiation of the Γ series is justified by the same locally uniform bounds as above.

Finally,

$$\int_0^L e^{\frac{q}{2}} \cos(\omega q) dq = \frac{\sqrt{x} - 1}{2(\omega^2 + \frac{1}{4})}, \quad \int_L^\infty \rho_1(q) \cos(\omega q) dq = \sum_{k \geq 1} \frac{\lambda_k x^{-\lambda_k}}{\lambda_k^2 + \omega^2}. \quad (152)$$

Thus the last line of (98), together with the rational difference $a_1 - a_\Gamma$, equals $(\frac{2}{L})(\mathfrak{a}'_x - A')$. This proves its diagonal is $a_\Gamma(\omega) + (\frac{2}{L})\mathfrak{a}'_x(\omega)$, as required. The calculation applies also at $\omega = 0$.

At zero frequency the result reads explicitly

$$\begin{aligned} (H_N)_{00} = a_0 - \left(\frac{2}{L}\right) \sum_{1 < n < x} \frac{\Lambda(n)}{\sqrt{n}} \log\left(\frac{x}{n}\right) \\ + \left(\frac{8}{L}\right)(\sqrt{x} - 1) + \left(\frac{2}{L}\right) \sum_{k \geq 1} \frac{1 - x^{-\lambda_k}}{\lambda_k^2}. \end{aligned} \quad (153)$$

Appendix D. Formal verification

Parts of the construction have counterparts checked in Lean with mathlib: the finite Euler and Möbius identities; the finite-window critical completion, domain-defined current and absorbing range; its finite current-form realization; the full simple-even ground construction and entire real-zero characteristic; and the bounded critical half-line operator with nonclosability of the uncompleted arithmetic operator and failure of a positive lower bound for its completion. The accompanying verification supplement, `verification/THEOREM_MAP.md`, identifies the precise hypotheses, compiled declarations, source hashes and reproducible build receipts for this coverage.

The holomorphic family, full-line Gram stationarity, causal leakage identity and trace-class support limit are proved in this article. They are outside the checked coverage just described, as are the explicit source-inverse formula and quantitative estimate, signed-current centering, alternative arithmetic interpolation with its diagonal repair, and zero-retaining entire-limit criterion. The supplement distinguishes these written proofs from their formally checked prerequisites.

Acknowledgements. ChatGPT participated extensively in the development of this work.

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